

SEARLE FIELD, NE, USA

WMO: 725621

Lat: 41.119N

Lon: 101.768W

Elev: 3250

StdP: 13.05

Time Zone: -7.00 (NAM)

Period: 97-19

WBAN: 94063

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.4	3.4	-8.1	4.0	-0.1	-2.4	5.4	6.2	33.9	33.0	30.4	33.9	6.3	260	0.603

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.5	97.2	67.2	93.5	66.9	90.7	66.5	72.0	86.2	70.7	85.0	69.3	83.6	11.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.2	117.2	77.2	66.4	109.6	76.2	64.5	102.7	74.9	38.0	86.4	36.6	84.8	35.5	83.9	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.9	25.5	21.6	DB	-13.5	101.8	3.9	3.1	-16.3	104.0	-18.6	105.9	-20.8	107.6	-23.7	109.9
			WB	-14.0	75.1	3.8	2.0	-16.8	76.5	-19.0	77.7	-21.2	78.8	-23.9	80.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	28.3	30.1	39.9	48.0	57.9	69.2	75.6	72.6	64.3	49.9	38.0	28.1
	DBStd	19.00	10.29	10.83	10.07	9.02	8.59	7.10	5.18	5.45	8.17	9.11	9.48	10.23
	HDD50	2894	673	558	332	140	27	1	0	0	4	111	368	680
	HDD65	6243	1138	978	777	511	250	38	4	10	111	472	809	1145
	CDD50	2994	0	1	20	80	272	578	795	699	433	108	8	0
	CDD65	867	0	0	0	1	30	166	333	244	90	3	0	0
	CDH74	11287	0	1	45	135	609	2148	3977	2737	1402	224	9	0
	CDH80	5235	0	0	6	26	210	959	2074	1283	618	59	0	0
	WSAvg	8.7	8.5	9.1	9.9	11.1	9.9	9.0	7.5	6.9	7.5	8.4	8.2	8.6
	PrecAvg	19.5	0.4	0.5	1.0	2.2	3.0	3.3	3.0	2.2	1.7	1.4	0.6	0.3
Precipitation	PrecMax	26.4	1.1	1.8	2.5	4.5	6.9	7.8	5.6	5.0	4.8	3.2	2.2	1.6
	PrecMin	6.9	0.0	0.0	0.1	0.0	0.7	0.0	0.9	0.6	0.3	0.1	0.0	0.0
	PrecStd	4.0	0.3	0.4	0.7	1.3	1.6	1.8	1.1	1.2	0.9	0.9	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.9	67.8	80.5	83.5	91.0	99.2	101.4	99.0	96.8	86.3	74.6	62.5
		MCWB	44.2	47.4	52.7	56.0	61.6	65.4	66.6	66.5	64.1	58.2	51.4	45.6
	2%	DB	55.2	59.6	72.5	78.6	86.1	94.1	98.3	94.8	91.4	80.8	68.1	54.7
		MCWB	41.2	43.0	49.7	54.0	60.7	66.2	68.2	67.0	63.8	56.8	48.9	41.4
	5%	DB	49.9	53.7	65.6	72.4	81.1	90.1	94.8	91.0	87.8	74.5	62.5	48.8
		MCWB	39.0	41.0	47.9	52.5	59.5	65.4	68.1	66.9	63.2	54.6	46.6	38.2
	10%	DB	44.6	47.4	59.3	66.4	75.9	85.7	90.9	87.8	82.2	68.4	55.1	43.6
		MCWB	35.9	37.7	44.7	50.3	57.9	64.9	68.1	67.0	61.8	52.2	43.2	35.5
	0.4%	WB	45.7	47.8	55.7	58.7	66.6	72.3	74.4	73.9	70.0	62.0	53.4	46.2
		MCDB	59.1	63.6	70.8	74.1	81.3	85.5	87.9	86.9	83.9	78.9	72.0	60.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	41.9	44.2	51.8	55.8	63.9	69.9	72.4	71.5	67.6	58.7	49.9	41.9
		MCDB	54.8	57.7	68.5	71.5	77.3	84.0	87.2	85.0	81.9	73.2	66.7	53.4
	5%	WB	39.2	41.2	48.9	53.6	61.9	68.3	71.1	70.1	65.9	56.2	47.0	38.9
		MCDB	48.4	52.8	63.7	69.6	75.5	82.9	86.3	83.4	80.6	71.0	60.7	48.3
	10%	WB	36.4	38.1	46.0	51.4	59.8	66.5	69.9	68.8	64.0	53.8	43.9	35.8
		MCDB	43.8	46.9	58.9	65.2	73.0	81.3	84.9	81.8	77.9	66.4	55.1	43.5
Mean Daily Temperature Range	5% DB	MDBR	25.7	25.3	28.2	26.5	26.1	27.1	27.5	27.0	29.7	29.2	28.4	25.0
		MCDBR	34.9	36.5	41.8	38.5	34.7	34.0	34.2	33.4	37.4	41.1	39.8	33.7
		MCWBR	22.2	21.9	21.7	18.0	14.7	11.6	10.8	11.7	13.8	19.1	22.2	21.6
	5% WB	MCDBR	32.4	34.9	37.5	33.8	28.5	28.3	28.0	27.3	29.9	35.1	38.3	32.1
		MCWBR	21.2	21.5	20.4	16.9	13.4	11.9	10.7	11.4	12.8	17.7	21.7	21.1
Clear-Sky Solar Irradiance	taub		0.257	0.265	0.290	0.315	0.341	0.356	0.381	0.372	0.335	0.295	0.266	0.257
	taud		2.554	2.531	2.486	2.432	2.432	2.419	2.393	2.404	2.472	2.554	2.557	2.535
	Ebn at Noon		293	306	307	304	296	291	282	282	287	290	287	284
	Edh at Noon		22	26	31	35	36	36	37	36	31	25	22	21
All-Sky Solar Radiation	RadAvg		744	1015	1399	1722	1999	2302	2270	1976	1607	1132	809	639
	RadStd		49	64	103	95	144	129	82	87	70	103	48	38

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page